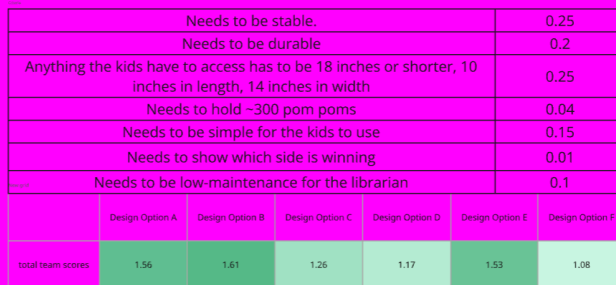


Problem: We were tasked with creating a voting machine for a middle school library.



1. A balance scale that tips to show which side has more pom poms.
2. Pom poms will slide into the jars down a slide similar to something found in a gumball machine.
3. The scale will have a perpendicular arrow attached to its center that will point at the angle of the winning choice.
4. An indicator card could be placed on the front of each jar to show a scale containing the pom poms. As the Pom Poms are put into the jars, the indicator card will flip up or down.
5. As either side of the scale goes down, it'll flip up an indicator to show how much each vote is winning or losing.
6. Again using the weight idea we use a gear ratio to make an arrow point at the winning decision.
7. Have a pom launcher. Two launchers each with a button where you put the pom into the corresponding launcher and press the button.
8. Have a tree with a lot of the signs. It'll be a pillar attached to the base of the scale and have two branches at a 120 degree angle. It's basically a glorified Y shape.



Orthographic Drawing

We worried about the scale being fragile. It ended up being pretty durable and just needs some smoothening out for joints. 300 mini pom poms can fit in the quart sized mason jars easily.

[First CAD Model \(click google drive Icon above\)](#)



First Prototypes (oldest to newest)
The Final Version will be 3d Printed!!!!